

A56 FindingSRAM

GAN CHIP — Deep Learning Accelerator (DLA) Engine

Schematic Review Video:

https://drive.google.com/file/d/1JfOFYj1IsYQqmhWTVg9je3te7nSkuVEE/view?usp=drive_link

Discord name	Affiliation	Experience	Role
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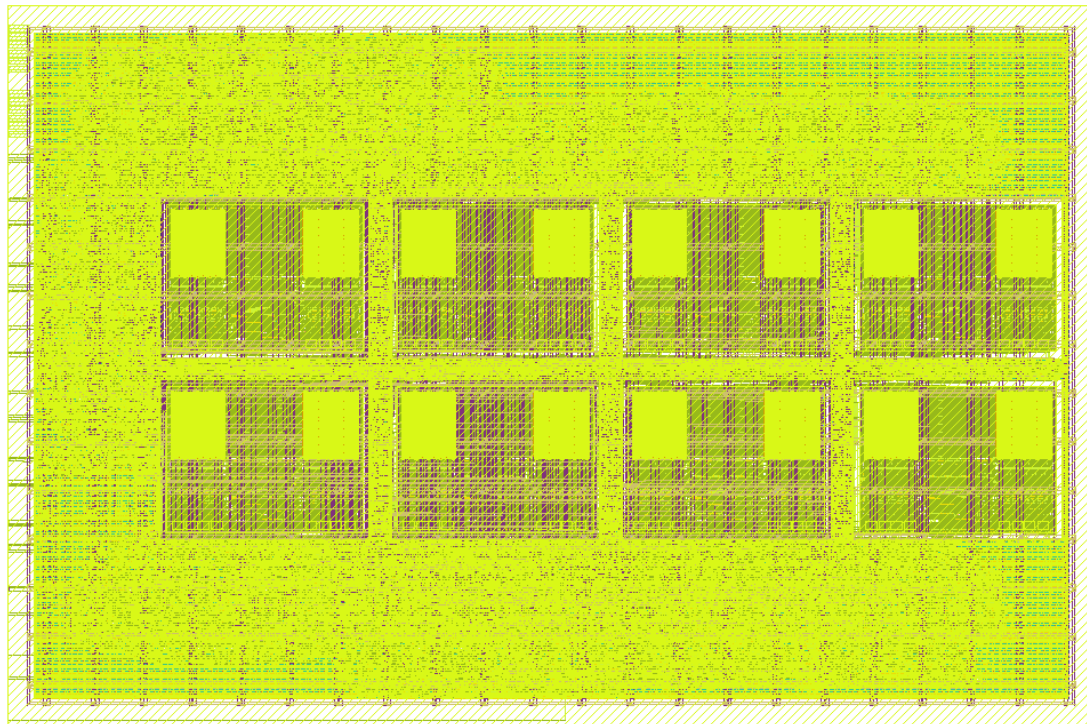
Results - Signoff Table

Check	Tool	Result
DRC (full)	Magic	0 violations COUNT: 0, illegal-overlap 0
DRC (routing)	OpenROAD DRT	0
Antenna	OpenROAD	0 nets / 0 pins closed via GRT diode margin 25 → 75
LVS	Netgen	Circuits match uniquely 25,342 devices / 14,058 nets, 0 errors
GDS XOR	Magic ↔ KLayout	0 differences
Setup	OpenSTA · 9 corners	MET - worst +14.98 ns @ max_ss_125C_3v00 · 0 vios
Hold	OpenSTA · 9 corners	MET - worst +0.117 ns @ min_ff_n40C_3v60 · 0 vios

Results - 9 Corner Timing

Corner	Setup WS (ns)	Hold WS (ns)	Setup vio	Hold vio
min_tt_025C_3v30	24.778	0.2135	0	0
nom_tt_025C_3v30	24.501	0.2144	0	0
max_tt_025C_3v30	24.178	0.2155	0	0
min_ss_125C_3v00	15.693	0.3648	0	0
nom_ss_125C_3v00	15.364	0.3660	0	0
max_ss_125C_3v00 ◀ worst setup	14.980	0.3674	0	0
min_ff_n40C_3v60 ◀ worst hold	29.283	0.1167	0	0
nom_ff_n40C_3v60	29.055	0.1174	0	0
max_ff_n40C_3v60	28.794	0.1182	0	0

Layout Render - Klayout

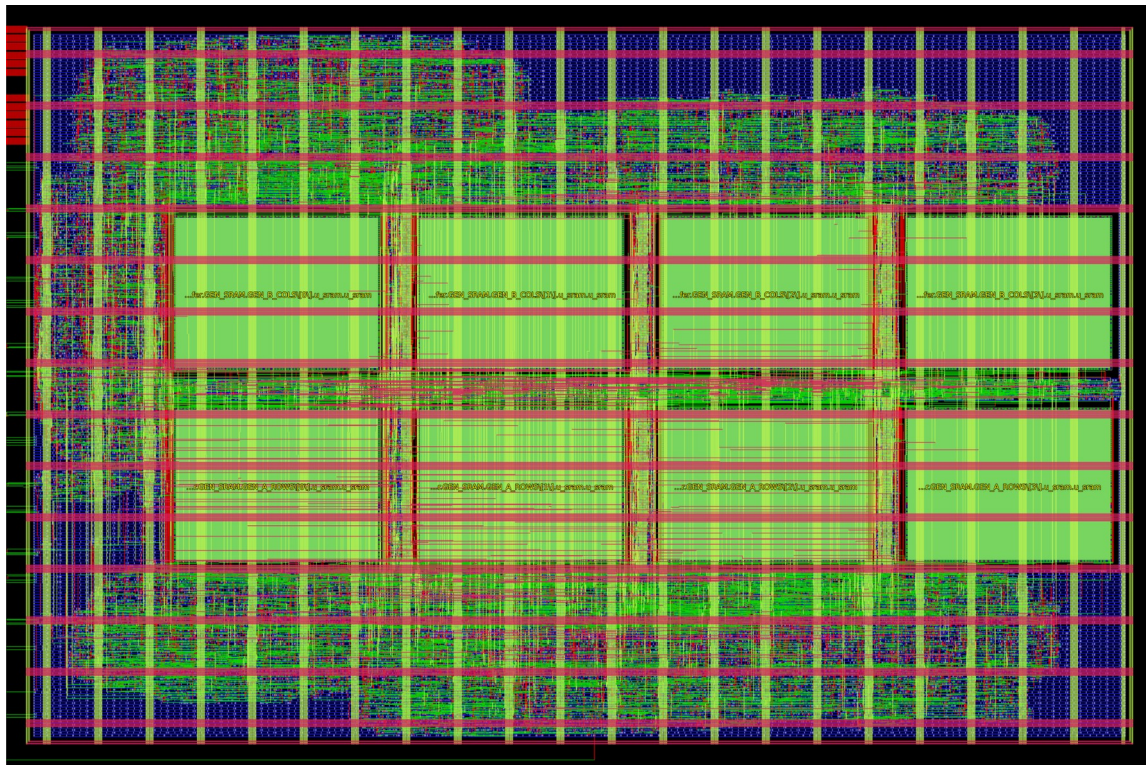


KLayout — GDS view. The streamed-out `dla_engine_chip.gds` (fill + every metal). Eight SRAM macros in a 4×2 cluster (shifted +130 μm east to open a west I/O channel); the yellow sea is std-cell logic + fill. DRC-clean, LVS-verified, antenna 0.

Floorplan at a glance.

- **A buffer** (weights, row broadcast) → bottom row, 4 macros
- **B buffer** (inputs, column broadcast) → top row, 4 macros
- **C buffer** (results) → flip-flops in the fabric — *no macro*
- Clustered 4×2 block (55 μm channels), shifted +130 μm east + perimeter logic/fill

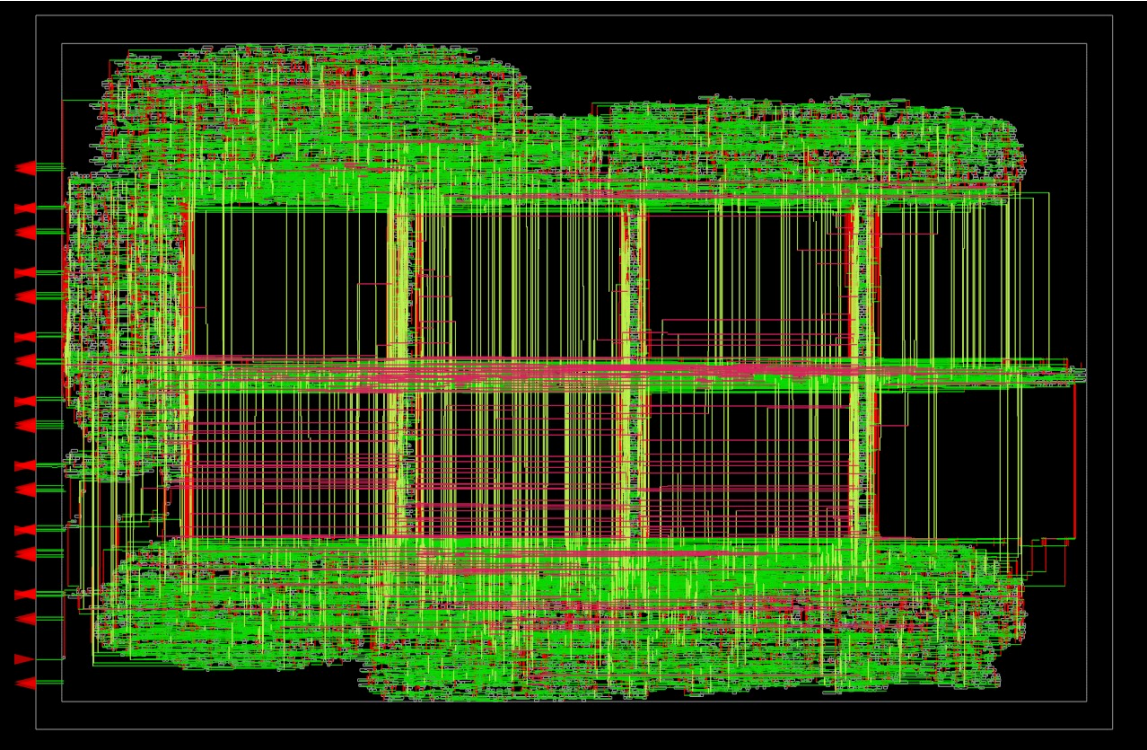
Layout Render - Openroad



OpenROAD — design view. Same die from the P&R database. Macros under u_dla.u_a_buffer / u_dla.u_b_buffer (top = B, bottom = A), pulled together so the SRAM → PE broadcast nets stay short; green diamonds are the boundary pins.

Signoff metrics	
Clock	25 MHz / 40 ns
Antenna	0 / 0
DRC / LVS	0 / match
Timing (9 corners)	met
Utilization	53.9 %
Total power (tt)	125 mW
Instances	83,862

Layout Render - Openroad



OpenROAD — signal nets only. Datapath + control routing (green/red/pink by layer) concentrated over the clustered macros and the mid-channel — the traffic that the clustering keeps short.

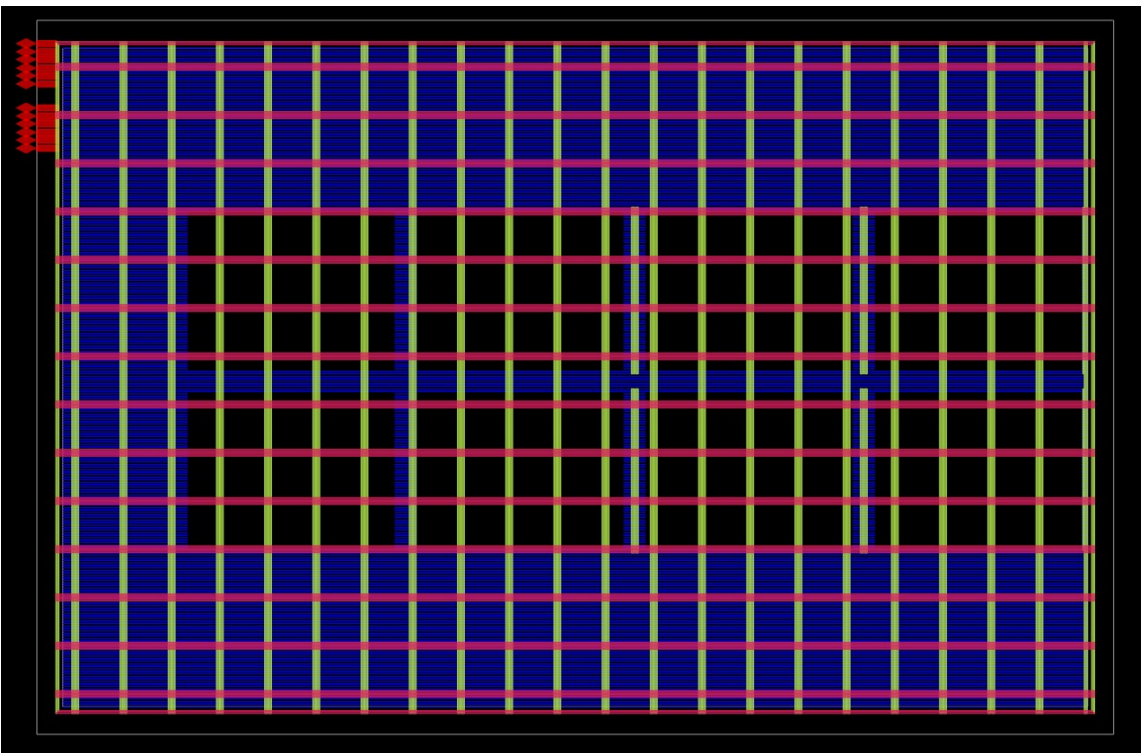
† Block IR drop computed by OpenROAD PSM on the power-connected design (RCX + SREF ~125 mW): worst rail sag DVDD 54.8 mV / DVSS bounce 70.6 mV $\Rightarrow \leq 3.8\%$ of 3.3 V, inside a typical 5% budget — the connector's via columns + M4/M5 core ring carry the load with margin. Full chip IR (through the shared padding) is an integration-stage analysis.

Signoff metrics	
Worst DVDD drop	54.8 mV (rail 3.245 V)
Worst DVSS bounce	70.6 mV
Average drop	48.6 / 62.1 mV (DVDD/DVSS)
Power-grid violations	0 · connectivity verified

How the die is built

Parameter	Value	Notes
Die / Core	1675 × 1110 μm 1,642,850 μm² core	ACV project block (1.859 mm²). ~40 μm die-to-core margin holds the PDN core ring.
Macros	8 × sram256x8m8wm1 301.30 × 224.93 μm ea.	Clustered 4×2 grid. x = 244.9 / 601.2 / 957.5 / 1313.8; y = 297.57 (A row) / 577.5 (B row). 55 μm channels, N orientation. Shifted +130 μm east to open a ~245 μm west channel for the serial bridge + west-edge I/O.
C buffer	flip-flops	16 × 24-bit registers + 1-cycle registered read. Removes the 9th macro; the register bank places freely in the fabric.
Target density	50 % → 53.9 % achieved	Antenna closed by the GRT diode-margin sweep (slide 11). Clustering shortens the broadcast nets; density 50 is the placement that routes them all clean.
Instances	83,862 total	33,315 std cells · 11,344 antenna diodes · 8 macros · balance = fill/tap/decap/endcap.
PDN	Metal4 ∥ + Metal5 ⊥ straps	Single 3.3 V DVDD/DVSS grid, 5 μm straps @ 75 μm pitch. Macro pins connect on Metal3 (OCD SRAM tops out at Metal3 — custom <code>pdn_cfg.tcl</code>). + a Metal4/Metal5 core ring in the 40 μm margin; the padframe's Metal2 DVDD/DVSS pins weld to it via <code>connect_power_v4.py</code> .
Ports	11 ACV pad terminals	9 signals (clk, rst_n, SCLK, MOSI, CS_N, MISO, busy, done, wb_done) + DVDD/DVSS on the WEST edge (W12–W22); the 53-bit DLA bus is wrapped behind a 4-wire serial bridge (62 pad-control ports).

Power delivery network

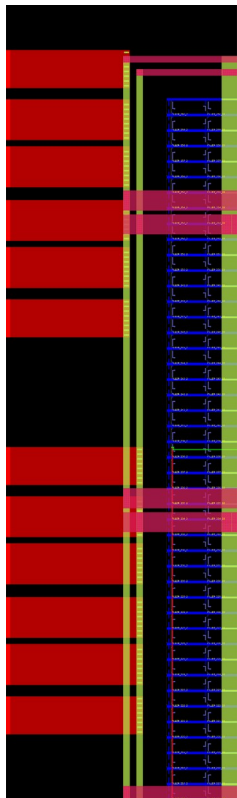


OpenROAD — power nets only. Vertical green = Metal4 DVDD/DVSS straps (5 μm @ 75 μm pitch); horizontal blue = per-row Metal1 rails. The dark rectangles are the 8 macros (fed on Metal3); green diamonds are the power pins.

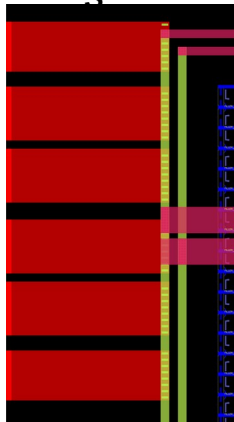
- **Single 3.3 V domain** (DVDD/DVSS) — no domain crossings, no level shifting. Running DVDD/DVSS as closely-spaced interleaved straps is intentional: it lowers PDN loop inductance and adds decoupling capacitance.
- **Both SRAM banks** tied to the grid explicitly on Metal3 via `PDN_MACRO_CONNECTIONS`; well/substrate taps in every AS std cell + macro (DRC-enforced spacing).
- **IR-drop & EM low: 125 mW at 25 MHz** with min-drive cells; straps sized far above demand. A Metal4/Metal5 core ring in the die-edge margin carries DVDD/DVSS:

Power delivery network - DEF template connection

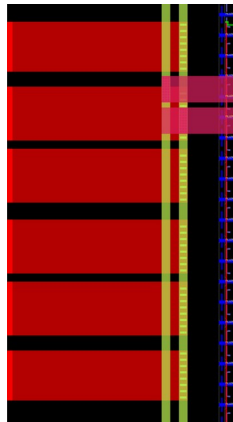
Power Pin
Connection



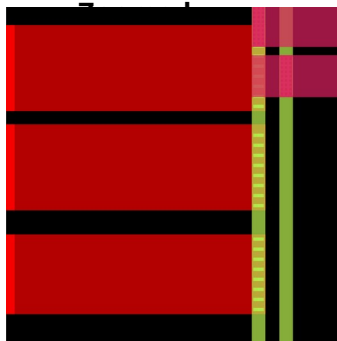
DVS
S



DVDD



DVSS



DVDD



`connect_power_v4.py` welds each padframe Metal2 DVDD/DVSS pin (west edge) to the core ring — a Metal2 reach + a via column per finger. DRC/LVS-clean, no connect-by-label.

Remaining Review Criteria

Analog / Matching

Fully digital — **no analog circuitry**. No PLL, no matched pairs, no sensitive nodes. The only analog-adjacent structures are the SRAM sense amps, pre-qualified foundry IP used as-is.

Reliability & PD Risk

- **Antenna: 0 violations** — closed to literal zero (slide 11); no residual charge-accumulation risk.
- **Latch-up:** taps in every std cell + macro, DRC-enforced spacing.
- **Voltage:** single 3.3 V, all AS cells + OCD SRAM 3.3 V-rated.
- **EM / current density:** negligible at 25 MHz, min-drive cells.
- **ESD:** pad-level clamps live in the auditor's padframe (pad cells are outside this project block).

Top-Level Integration

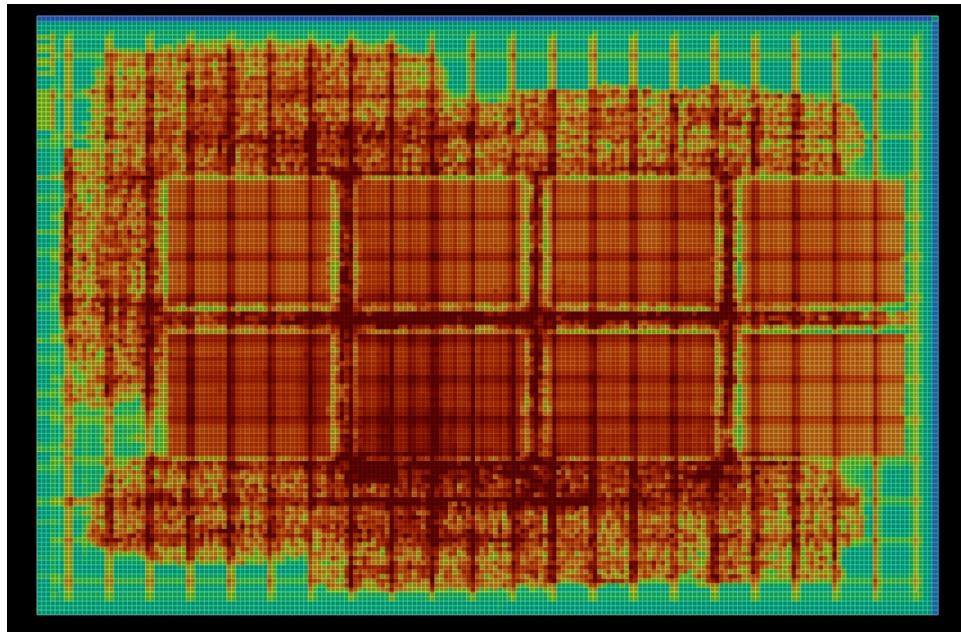
- Top cell **dla_engine_chip** everywhere (RTL ↔ netlist ↔ layout).
- Pin names match RTL 1:1 — proven end-to-end by LVS.
- Ports = pad-control terminals at the template positions; DVDD/DVSS welded to the core ring by connect_power_v4.py.
- Hardened against the auditor's A56_ACV.def padframe template; presents the 11 ACV pad terminals on the west edge.

Rubric 1 recap (DRC/LVS): Magic DRC 0, routing DRC 0, GDS XOR 0; Netgen LVS matches uniquely across 25,342 devices — layout corresponds to the RTL device-for-device. Full signoff on slide 2.

Antenna closure: 4 → 2 → 0

Experiment	Antenna	Note
clustered 4×2 baseline	low	clustering keeps the SRAM broadcast nets short (from longtin)
new west-edge template · margin 25	5	auditor moved every pin 100 μm → routing re-rolled (was 1)
GRT diode margin 50	3	cleared 2 of the 3 marginals
GRT diode margin 75	0	winner — cleared net5, the structural buffer → output net

Lesson: the clustered macros keep the base count low, but the auditor's new pin template re-rolled routing to 5 nets. Because the pin positions are auditor-fixed and can't be moved, a more aggressive global-route diode margin (25 → 75) is more targeted than a density re-roll — it cleared even net5, the one structurally-long net that recurs every harden. DRC/LVS stayed clean.



OpenROAD — placement-density heatmap. Red = packed, teal = sparse. The datapath logic packs tightly *around the clustered macros* (the placer pulls connected cells to the memories), with routable breathing room at the perimeter — exactly the compact arrangement that keeps the broadcast nets short.

UVM environment (pyuvm on cocotb)

UVM (IEEE 1800.2) builds a testbench from cooperating components that speak in **transactions**, not wires. Here it runs in Python via **pyuvm** (a faithful port of the standard classes) driving the DUT through Icarus — every class name is identical to SystemVerilog UVM.

DlaTest (uvm_test) — builds env, starts sequence, checks tally

DlaEnv (uvm_env) — wires analysis ports

DlaAgent (uvm_agent)

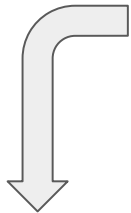
— Sequencer — routes DlaOp items

Driver — the only pin-wiggler

Monitor — passive; rebuilds C from pins

DlaScoreboard

golden matmul + checker



Sequence → Sequencer → Driver → (pins) → **DUT**

DUT → (pins) → Monitor → (observed C) → Scoreboard

Driver → (applied A,B) → Scoreboard

Mapped to the accelerator

- **DlaOp** = one matmul job (A 4×256 weights, B 256×4 inputs).
- **Sequence** = 1 directed (A=B=1 ⇒ every C=256) + 3 constrained-random ops.
- **Driver** = real protocol: stream A/B, pulse **start**, wait **wb_done**, read 16 C.
- **Monitor** = reconstructs the 16 C outputs from the read-back bus.
- **Scoreboard** = predicts C=A·B (INT8, 24-bit wrap) and compares.

Scope: this env black-box-verifies the DLA core (dla_engine_top), instantiated unchanged inside the submitted dla_engine_chip. The bridge-wrapped chip is additionally directed-tested at the ACV pads (tb/dla_engine_chip_tb.sv), bit-exact against the same goldens.

```
# cd /foss/designs/tb/uvm && make
[directed_ones] PASS: 16/16 match (C[0][0]=256)
[rand_0] PASS  [rand_1] PASS  [rand_2] PASS
ALL 4 TRANSACTIONS PASSED — 4 passed, 0 failed
```

Notes for the reviewer

Why C moved to flip-flops

C holds only 16 accumulators (48 bytes) — a wasteful 64×8 macro. As flip-flops it costs ~408 registers (tiny in an 85k-instance die), **removes a macro**, cuts writeback 48 → 16 cycles, and drops the C read latency 4–5 → 1 cycle. A/B stay SRAM because they need four parallel 8-bit lanes each.

Post-layout timing — no SDF simulation: the 3.3 V AS standard-cell library ships with no `specify` timing blocks (0 added in this design), so OpenROAD dynamic gate-level sim isn't possible in this open toolset. Post-layout timing is

signed off by SPEF-backed corner STA (the standard method) and functional equivalence by zero-delay GLS. The open PDKs 5-V library does carry specify (916 blocks) — so

it's a library, not a toolset limitation. Please

turn off Signal to get the Power/Ground view (slide 5)

```
# headless screenshot (full die), for the design views:
QT_QPA_PLATFORM=offscreen openroad -exit _or_render.tcl
# read_db ../final/odb/dla_engine_chip.odb ; save_image -area {0 0 1675 1110}
out.png
```

```
# the raw GDS stream is a KLayout job (OpenROAD has no GDS reader):
```

```
klayout runs/acv_ring4/final/gds/dla_engine_chip.gds
```

Why cluster the macros

A roomy die spreads the memories apart, so the SRAM → PE broadcast nets route long and trip antennas. Pulling the 8 macros into a compact central block (55 μm channels) shortens those nets at the source; a density sweep then finds the placement (d50) that routes the residue clean. Result: antenna 0, timing/DRC/LVS unchanged.